

Nominal Anchors and Real Consequences: Inflation Targeting, Debt, and Investment

preliminary

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The views expressed herein are those of the authors and not necessarily those of the Federal Reserve Bank of San Francisco, or the Federal Reserve System.

What we do

secular trends in macro-finance since the 1980s often explained by real factors (demography, productivity, capital risk)

we ask: can the *monetary policy regime* itself play a role?

develop GE connections between inflation targeting and wider historical trends:

1. decline in long-term real interest rates on nominal bonds
2. increase in private (nominal) debt relative to GDP
3. investment 'drought' (declining I/Y)
4. stability of risky real returns to capital
5. strengthening of inflation targeting since the 1980s ($\phi_\pi \uparrow$)

can the monetary policy regime (5) be a unified explanation of (1)–(4) in a GE setting?

credible inflation targeting $\implies$ lower inflation volatility

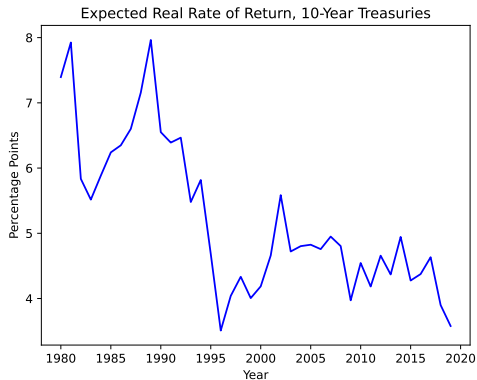
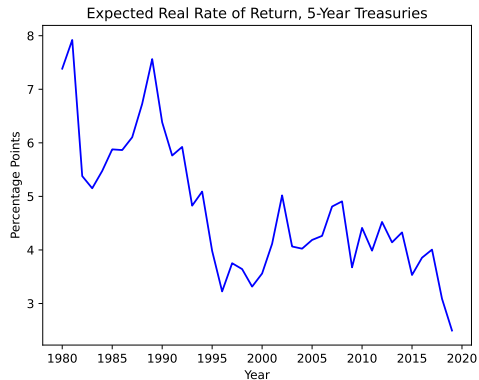
- nominal bonds become *safer* in real terms
- $\implies$ inflation risk premium + interest rate risk declines $\implies R_0^* \downarrow$
- $\implies$ savers supply more savings in nominal assets $\implies$ debt $\uparrow$
- $\implies$ capital becomes relatively *more* risky $\implies K \downarrow, \mathbb{E}_0[R_0^K] \uparrow$

successful nominal anchoring transforms the *risk characteristics* of nominal assets, triggering portfolio reallocation away from risky capital toward safer nominal bonds

1. Stylized facts since the 1980s
2. A three-period macro-finance model
3. Analytical results: zero debt stochastic steady state
4. Analytical results: positive (nominal) debt
5. Conclusion

Fact 1: Decline in long-term rates

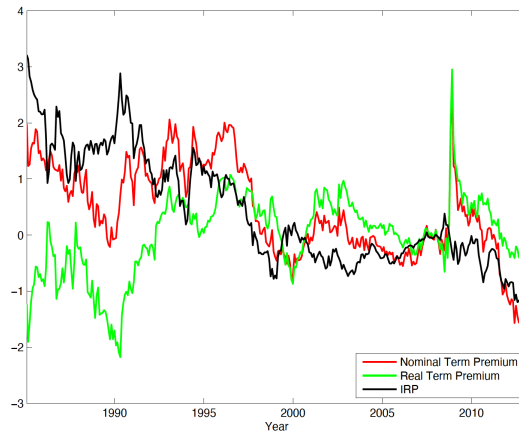
expected real yields on US nominal bonds



Fact 1: Decline in long-term rates

Inflation Risk Premium for the UK

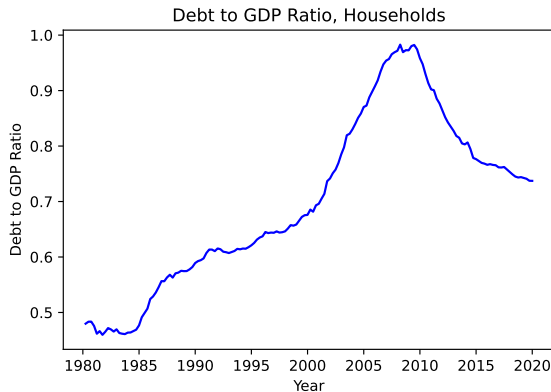
10-year Term Premium Comparison



Abrahams, Adrian, Crump, and Moench (2016) for the UK.

Fact 2: Rise in private (nominal) debt

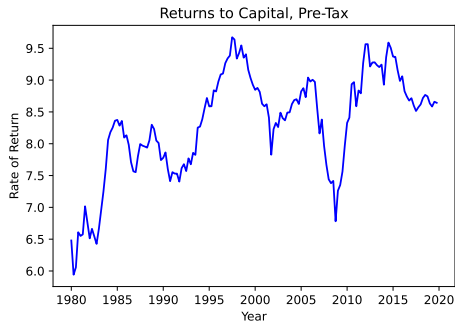
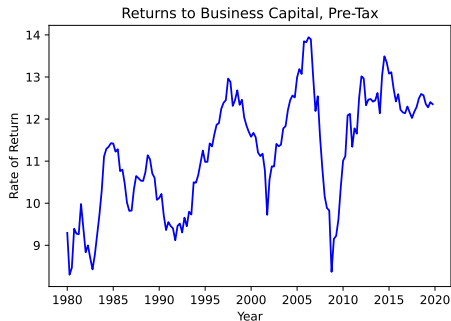
household debt relative to GDP in the US increased since 1980



Source: Our calculations based on Mian Straub & Sufi (2023)

Fact 3: Returns to capital stable or rising

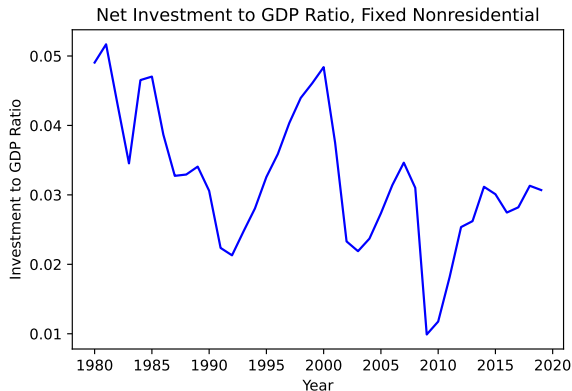
real return to capital, US data



Source: based on Gomme, Ravikumar, & Rupert (2019)

Fact 4: Investment drought

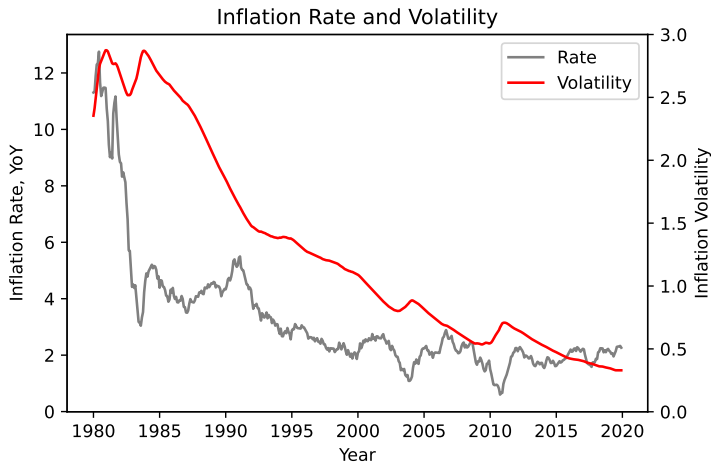
investment to GDP declined



Source: our calculations

Fact 5: Strengthening of inflation targeting

post Volcker, central banks adopt inflation targeting. US data (π , level and rolling sd)



Three Periods: ($t = 0, 1, 2$), flexible price economy

Agents

- Borrower and Savers
- Central bank (Taylor rule)

Assets

- Short-term nominal bond (zero supply)
- Long-term nominal bond
- Capital: illiquid, cannot sell at date 1

Endowments

- exogenous endowment at all dates

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Shocks

All uncertainty resolves at date 1

Aggregate Risk

- Preference shock ξ_1
- Source of inflation volatility
- inflation risk premium

Idiosyncratic Liquidity Risk on Savers

- Prob. λ : forced to liquidate at date 1
- Low income y_L in liquidity state
- interest rate risk on long-term bond

Two types $j \in \{b, s\}$: borrowers and savers, each mass one

- Borrowers: discount factor β_b^t , with β_b small (always constrained)
- Savers: discount factor $\beta^t \xi_t$, where ξ_t is a stochastic preference shock

Preference shock ξ_1 :

- $\xi_0 = 1$, $\xi_1 \sim [\xi_L, \xi_H]$ with $\mathbb{E}[\xi_1] = 1$, $\text{Var}(\xi_1) > 0$
- Date-2 persistence: $\xi_2 = \xi_1^\rho$ with $\rho \in (0, 1)$
- All uncertainty resolves at date 1

Idiosyncratic Liquidity Shock

Each saver draws $\eta \in \{0, 1\}$ (wt on date 2 utility) at beginning of date 1, i.i.d., independent of ξ_1 :

	Impatient ($\eta = 0$)	Patient ($\eta = 1$)
Probability	λ	$1 - \lambda$
Date-1 endowment	y_L	$y_H \geq y_L$
Long bonds at date 1	Sells at Q_1^L	Holds to maturity

Saver's date-0 utility:

$$U_s = u(C_{s,0}) + \beta \mathbb{E} \left[\xi_1 \left(\lambda u(C_{s,1}^I) + (1 - \lambda) \left[u(C_{s,1}^P) + \beta \xi_1^{p-1} u(C_{s,2}^P) \right] \right) \right]$$

$$u(C) = C^{1-\gamma}/(1-\gamma)$$

1. **Short-term nominal bond** — issued at date 0, pays $(1 + i_0^S)$ at date 1
 - Zero supply: pins short-term real interest rate
 - Real return: $(1 + i_0^S)/\Pi_1$ — exposed to date-1 inflation
2. **Long-term nominal bond** — issued at date 0 at price Q_0^L , pays 1 at date 2
 - Liquid: can be resold at date 1 at price $Q_1^L = \frac{1}{1+i_1^S}$
 - Impatient type sells at $Q_1^L \Rightarrow$ exposed to both inflation risk and interest rate risk
3. **Real capital K** — invested at date 0, produces $\bar{A}K^\alpha$ at date 2
 - Illiquid: cannot be sold or pledged at date 1
 - No inflation exposure

Saver's Budget Constraints

Date 0 (before learning η):

$$P_0 C_{s,0} + P_0 K + Q_0^L b^L = P_0 Y_0$$

Date 1, Impatient type (λ): sells all long bonds, consumes everything

$$P_1 C_{s,1}^I = P_1 y_L + Q_1^L b^L$$

Date 1, Patient type ($1 - \lambda$): receives y_H , holds bonds to maturity, saves s_1

$$P_1 C_{s,1}^P + \frac{\lambda}{1 - \lambda} Q_1^L b^L = P_1 y_H$$

$$P_2 C_{s,2}^P = P_2 \bar{A} K^\alpha + P_2 Y_2 + \frac{b^L}{(1 - \lambda)}$$

Issue long-term nominal bonds, face debt limit

$$b^L/P_0 \leq d_0/R_0^{L*}$$

- Consume at dates 0 and 2 only:

$$P_0 C_{b,0} = P_0 Y_{b,0} + Q_0^L b^L, \quad P_2 C_{b,2} = P_2 Y_{b,2} - b^L$$

- Always constrained (β_b small)

Taylor rule:

$$1 + i_1^S = \frac{1}{\beta} \Pi_1^{\phi_\pi}, \quad \phi_\pi > 1$$

- policy regime characterized by $\phi_\pi > 1$
- normalize $P_0 = 1$
- Assume $\Pi_2 = \Pi_1^\delta$; $\delta \geq 0$ (Krugman 1998 special case: $\delta = 0$)

Date-1 long bond price:

$$Q_1^L = \frac{1}{1 + i_1^S} = \beta \Pi_1^{-\phi_\pi}$$

⇒ Links monetary policy to liquidity risk: Q_1^L fluctuates with ξ_1 through Π_1

⇒ More aggressive targeting stabilizes the resale price of long bonds

Three pricing equations:

- **Short bond** (no supply):

$$1 = \mathbb{E} \left[\left(\lambda M'_{01} + (1 - \lambda) M^P_{01} \right) \frac{1 + i_0^S}{\Pi_1} \right]$$

- **Long bond** (hybrid: impatient sells at date 1, patient holds to date 2):

$$Q_0^L = \mathbb{E} \left[\lambda M'_{01} \frac{Q_1^L}{P_1} + (1 - \lambda) M^P_{02} \frac{1}{P_2} \right]$$

- **Capital** (only patient types benefit, liquidity tax $(1 - \lambda)$):

$$1 = (1 - \lambda) \mathbb{E} \left[M^P_{02} R_K \right], \quad R_K = \alpha \bar{A} K^{\alpha-1}$$

allocations, prices, interest rates such that

1. households optimize,
2. bond markets clear, and
3. the Taylor rule holds

Zero-debt benchmark

Set $d_0 = 0$: no borrowing or lending. savers only save in capital.

from patient type's date-1 Euler + Taylor rule, solve for equilibrium inflation:

$$\Pi_1 = \bar{D} \xi_1^a, \quad \text{where } a \equiv \frac{1 - \rho}{\phi_\pi - \delta}, \quad \bar{D} \equiv \left(\frac{Y_2}{y_H} \right)^{\gamma/(\phi_\pi - \delta)}$$

inflation variance:

$$\text{Var}(\log \Pi_1) = \left(\frac{1 - \rho}{\phi_\pi - \delta} \right)^2 \text{Var}(\log \xi_1) \quad \Rightarrow \quad \text{strictly decreasing in } \phi_\pi$$

cumulative price level: $P_2 = \Pi_1 \Pi_2 = \Pi_1^{1+\delta}$

$$\text{Var}(\log P_2) = (1 + \delta)^2 a^2 \text{Var}(\log \xi_1) \quad \text{amplified by persistence } \delta$$

Zero-debt: short-bond inflation risk premium

expected real return on the short nominal bond:

$$\bar{R}_0^{S*} = \bar{R}_1^f \cdot \underbrace{\frac{\mathbb{E}[\xi_1^{-a}]}{\mathbb{E}[\xi_1^{1-a}]}}_{\text{IRP}^S > 1}$$

- $\uparrow \phi_\pi \implies \downarrow \bar{R}_0^{S*}$ (lower compensation for inflation risk)
- as $\phi_\pi \rightarrow \infty$: $\text{IRP}^S \rightarrow 1$. Perfect IT eliminates inflation risk premium
- IRP^S independent of γ , λ , or ω

Decomposing the real bond yield

use bond pricing equation $1 = \mathbb{E}[M \cdot (1 + i_0^S)/\Pi_1]$:

$$R_0^{S*} = \underbrace{\frac{1}{\mathbb{E}[M_{01}]}_{R^f} + \underbrace{\left(-\frac{\text{Cov}(M_{01}, (1 + i_0^S)/\Pi_1)}{\mathbb{E}[M_{01}]} \right)}_{\text{inflation risk premium } \psi}$$

at $b^L = 0$: $C_{s,1}$ is deterministic for each type $\implies R^f = \bar{R}_1^f$ is invariant to ϕ_π

$$\uparrow \phi_\pi \implies \downarrow \psi \implies \downarrow R_0^{S*}$$

Why does ϕ_π lower the risk premium?

bonds as bad hedges:

- high ξ_1 = high marginal utility **and** high inflation
- $\implies$ nominal bonds pay poorly when consumption is most valuable
- $\implies$ positive inflation risk premium

but *which component* of the covariance responds to ϕ_π ?

Channel	Depends on ϕ_π ?	Contribution
Risk-free rate $\bar{R}_1^f$	No	0%
SDF volatility $\sigma_M \propto \sigma$	No	0%
Correlation $\text{Corr}(M, \Pi_1^{-1}) = -1$	No	0%
Inflation vol $\sigma_{\Pi^{-1}} \propto a \sigma$	Yes	100%

safe asset demand story: $\uparrow \phi_\pi$ reduces volatility of real bond payoffs $\implies \bar{R}_0^{S*} \downarrow$
bonds remain bad hedges (correlation = -1)

Term structure of inflation risk premia

Short-bond IRP:
$$\text{IRP}^S = \frac{\mathbb{E}[\xi_1^{-a}]}{\mathbb{E}[\xi_1^{1-a}]}$$

Long-bond IRP (patient, hold-to-maturity):
$$\text{IRP}^L = \frac{\mathbb{E}[\xi_1^{-a(1+\delta)}] \mathbb{E}[\xi_1^\rho]}{\mathbb{E}[\xi_1^{\rho-a(1+\delta)})]}$$

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Define $\Phi(\delta) \equiv \log \text{IRP}^L - \log \text{IRP}^S$:

- (i) $\Phi(0) < 0$: long bond has *smaller* IRP
- (ii) $\Phi'(\delta) > 0$: long-bond IRP strictly increasing in persistence
- (iii) $\exists! \delta^*$: for $\delta > \delta^*$, standard ordering $\text{IRP}^L > \text{IRP}^S$
- (iv) for $\delta > \delta^*$: $d\Phi/d\phi_\pi < 0$ — term premium compresses in ϕ_π

Zero-debt: capital is independent of ϕ_π

capital Euler (only patient types benefit; liquidity tax $1 - \lambda$):

$$1 = (1 - \lambda) \mathbb{E} \left[M_{02}^P R_K \right], \quad R_K = \alpha \bar{A} K^{\alpha-1}$$

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at $b^L = 0$: $M_{02}^P = \beta^2 \xi_1^\rho (Y_2 + \bar{A} K^\alpha / C_{s,0})^{-\gamma}$

- R_K is independent of ϕ_π : capital has no inflation exposure
- R_K is increasing in λ : liquidity tax requires higher compensation
- R_K is independent of $\omega = \left(\frac{y_H}{y_L}\right)^\gamma \geq 1$: capital priced by patient type only

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Illiquidity premium: $R_K - \bar{R}_0^{L*} > 0$

- R_K invariant to ϕ_π , while $\bar{R}_0^{L*}$ strictly decreasing
- $\implies$ **spread widens** as inflation targeting becomes more aggressive
- R_K independent of ω , but bond price increasing in $\omega \implies$ income risk amplifies

At zero debt, more aggressive inflation targeting ($\uparrow \phi_\pi$):

1. $\bar{R}_0^{S*} \downarrow$: short-bond IRP falls (inflation volatility channel, 100%)
2. $\bar{R}_0^{L*} \downarrow$: long-bond IRP falls (same mechanism)
3. **term premium compresses** (when $\delta > \delta^*$)
4. R_K unchanged: capital return pinned by real fundamentals
5. **illiquidity premium $R_K - \bar{R}_0^{L*}$ widens**

Next: what happens when we turn on debt ($b^L > 0$)?

Positive debt: perturbation around stochastic steady state

expand around the **stochastic** zero-debt equilibrium (not a deterministic SS)

- inflation is stochastic even at $b^L = 0$
- $\implies$ nonzero inflation risk premium embedded from the outset
- contrasts with standard perturbation where risk premia vanish at first order

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at $b^L > 0$: impatient type's consumption becomes **stochastic**

$$C'_{s,1} = y_L + \frac{Q_1^L}{P_1} b^L = y_L + \beta \Pi_1^{-(1+\phi_\pi)} b^L$$

- $C'_{s,1}$ depends on Π_1 through liquidation payoff $\propto \Pi_1^{-(1+\phi_\pi)}$
- low base $y_L \implies$ marginal utility amplified by $\omega = (y_H/y_L)^\gamma$
- M'_{01} now covaries with inflation $\implies$ interest rate risk channel activated

Proposition. At $b^L > 0$, $\partial \bar{R}_0^{L*} / \partial \phi_\pi < 0$ through two channels:

(i) Inflation risk channel (present for all λ):

- $\uparrow \phi_\pi$ reduces $\text{Var}(\log \Pi_1) \implies$ compresses IRP
- same mechanism as zero-debt benchmark

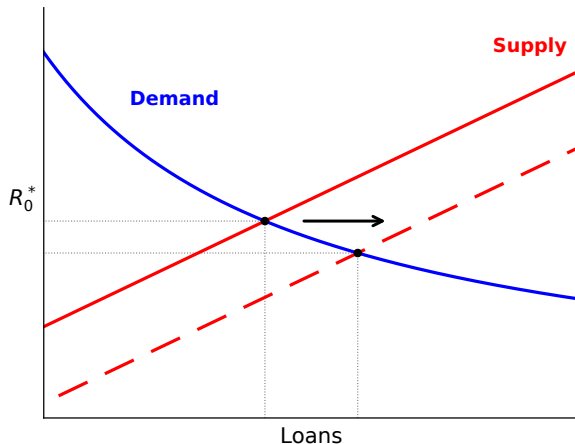
(ii) Interest rate risk channel ($\propto \lambda$, amplified by ω):

- $\uparrow \phi_\pi$ stabilizes $Q_1^L = \beta \Pi_1^{-\phi_\pi} \implies$ reduces capital-gain risk
- priced by impatient type's elevated SDF (ω times patient type's)
- vanishes at $\lambda = 0$; strengthens in λ and ω

Intuition: supply and demand for credit

more aggressive inflation targeting $\Rightarrow$ lower R_0^{L*}

$\Rightarrow$ more nominal debt supplied by savers $\Rightarrow$ lower R_0^L , more credit in equilibrium



[binding debt limit means the demand curve is $R_0^{L*} = d_0/\text{Loans}$]

Crowd-out: the question

when bonds become safer, do savers **displace capital** or just **save more**?

write total savings as $S_0 \equiv Y_0 - C_{s,0} = K + Q_0^L b^L$

differentiate with respect to bond holdings b :

$$\left. \frac{dK}{db} \right|_{b=0} = \underbrace{-1}_{\text{portfolio margin}} + \underbrace{\left. \frac{dS_0}{db} \right|_{b=0}}_{\text{savings margin}}$$

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- **portfolio margin** = -1 : holding S_0 fixed, each dollar of bonds displaces a dollar of capital
- **savings margin** = $1 + k$: when bonds are more attractive, saver may cut $C_{s,0}$ and save more
- $k \equiv dK/db$ is the **crowd-out coefficient**

Crowd-out: three forces

linearize capital Euler $(1 - \lambda) \mathbb{E}[M_{02}^P R_K] = 1$ around $b = 0$ (stochastic):

$$0 = \underbrace{(1 + k) Q_0^L \cdot \frac{\gamma}{\bar{C}_0}}_{\text{(a) date-0 smoothing}} + \underbrace{k \cdot \frac{(1 - \alpha)}{\bar{K}_1}}_{\text{(b) diminishing returns}} + \underbrace{k \cdot \frac{\gamma R_K}{\bar{C}_2}}_{\text{(c) date-2 smoothing}}$$

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(a) Date-0 consumption smoothing (drives crowd-out)

- saver saves more ($1 + k > 0$) $\implies$ cuts $C_{s,0} \implies$ higher marginal utility today
- resists further saving $\implies$ limits the savings margin

(b) Diminishing returns (absorbs displacement)

- capital falls ($k < 0$) $\implies$ marginal product R_K rises $\implies$ partially stabilizes capital

(c) Date-2 consumption smoothing (limits crowd-out)

- capital falls $\implies \bar{C}_2 = \bar{A}K^\alpha + Y_2$ falls $\implies$ higher marginal utility at date 2
- makes remaining capital more valuable $\implies$ resists further displacement

Capital extension: implications

$\uparrow \phi_\pi \implies \text{safer nominal bonds} \implies b \uparrow \implies K \downarrow \text{ (one-for-one) } \implies \mathbb{E}_0[R_0^K] \uparrow$

mechanism: improved risk-return profile of nominal bonds crowds out capital investment

Capital extension: implications

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Equilibrium implications ($\uparrow \phi_\pi$):

- $\bar{R}_0^{L*} \downarrow$ (declining real rate on nominal bonds)
- debt $\uparrow$ (rising private credit)
- $K \downarrow$ (investment drought)
- R_K stable (pinned by real fundamentals)
- $R_K - \bar{R}_0^{L*} \uparrow$ (widening safe-risky spread)
- term premium $\downarrow$ (when $\delta > \delta^*$)

aggressive inflation targeting regime:

- reduces inflation volatility $\implies$ lowers R_0^*
- low inflation vol & interest rate risk $\implies$ nominal bonds safer $\implies$ supply of savings in nominal assets goes up
- capital becomes relatively *more* risky $\implies$ supply of savings in real assets falls
- equilibrium: $\uparrow$ private credit, $\downarrow$ natural rate on nominal bonds, $\downarrow$ investment

a unified account of macro-finance trends since the 1980s:

declining real rates, rising debt, investment drought, stable returns on capital
all as equilibrium responses to a single shift in monetary policy regime

What next

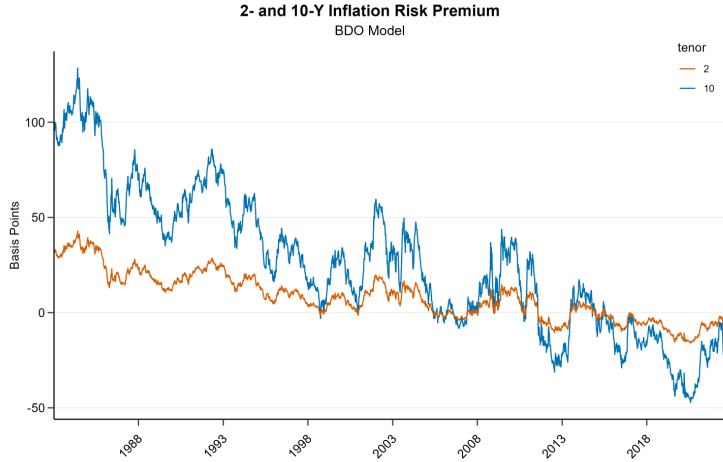
quantitative exercise:

- infinite horizon model with aggregate uncertainty
- decompose the contribution to decline in r^* from ϕ_π vs other forces (demography, productivity, capital risk)

THANK YOU!

Fact 1: Decline in long-term rates

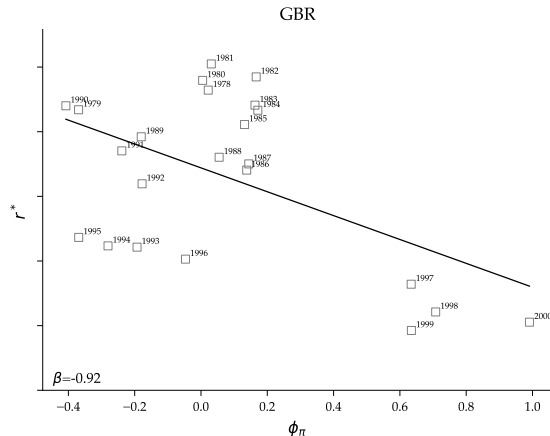
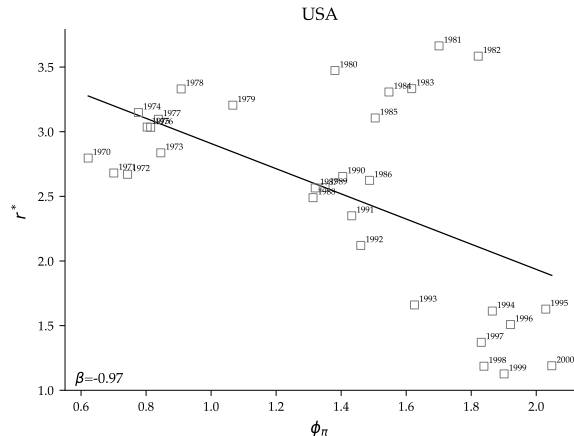
Inflation Risk Premium for the US



Breach, D'Amico, Orphanides (2020) for the US.

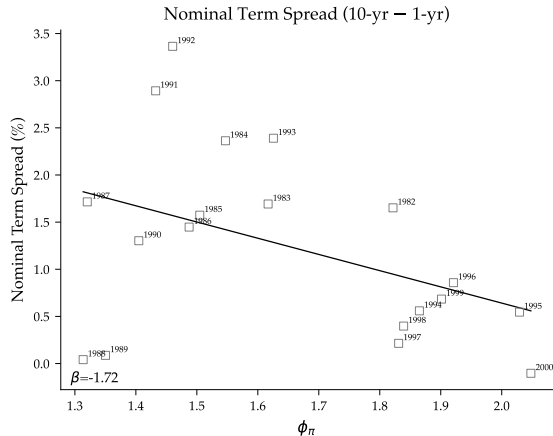
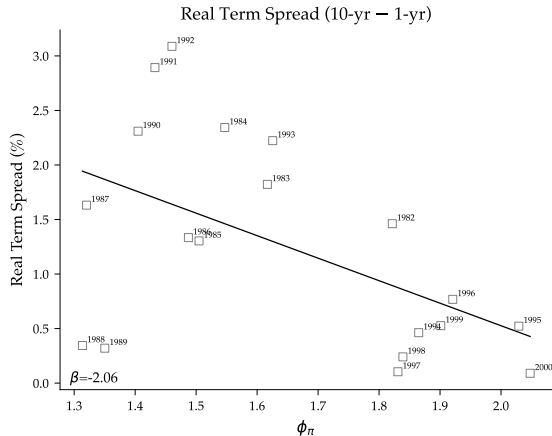
Monetary policy and r-star

HLW estimates of r^* and our estimates of ϕ_π based on Carvalho Nechio Tristao (2021)



Monetary policy and term spread

Term spread (10 y - 1 y) + our estimates of ϕ_π based on Carvalho Nechio Tristao (2021)



Monetary policy and DGGT r-star spread

DGGT term spread + our estimates of ϕ_π based on Carvalho Nechio Tristao (2021)

